Signals and Systems Lecture-13: Sampling Theorem

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Sampling Introduction (1)

Consider a continuous-time signal $x(t)$.

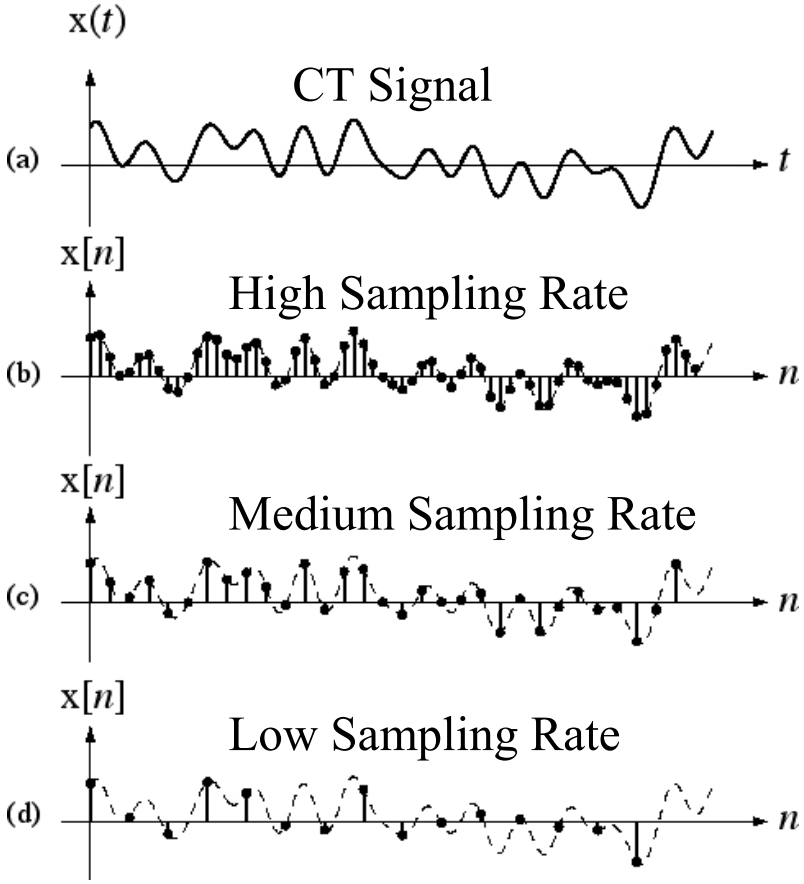
- Sampling $x(t)$ at a sampling frequency f_s corresponds to taking snapshots of $x(t)$ every $T_s = 1/f_s$ seconds.

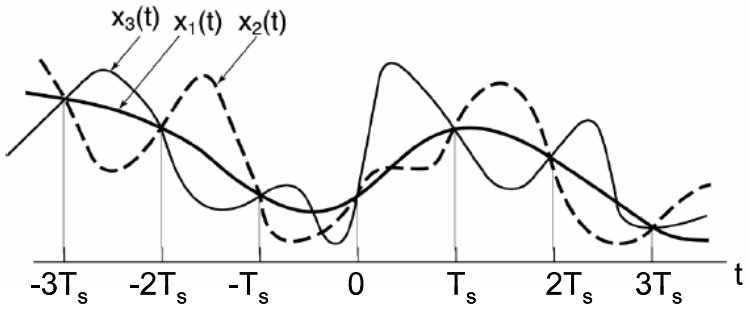
- where:
  - $x(t)$ is an **analog** signal: It has a continuous time and a continuous amplitude.
  - $x[n]$ for $n \in \mathbb{Z}$ are the regularly spaced samples.

$$x[n] = x(nT_s) \;\; ; \;\; n \in \mathbb{Z}$$

  - At the output of a sampler, the signal will have a discrete-time and a continuous amplitude.

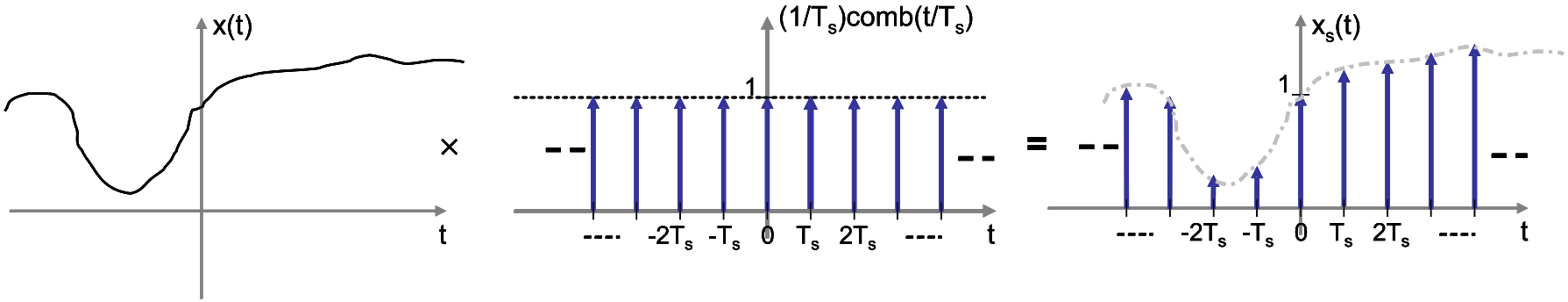
Sampling Introduction (2)





- Lots of signals have the same samples.
- The sampling theorem answers the question: under what conditions can the original continuous-time signal be reconstructed from its samples?

Impulse Sampling (1)



- Impulse sampling can be performed by multiplying the signal $x(t)$ by the sampling function:

$$p(t) = \sum_{n\in\mathbb{Z}} \delta(t - nT_s) = \frac{1}{T_s}\mathrm{comb}\left(\frac{t}{T_s}\right)$$

- In this case, the sampled signal is given by:

$$x_s(t) = x(t)p(t) = x(t)\sum_{n\in\mathbb{Z}} \delta(t - nT_s)$$

$$= \sum_{n\in\mathbb{Z}} x(t)\delta(t - nT_s) = \sum_{n\in\mathbb{Z}} x(nT_s)\delta(t - nT_s)$$

Impulse Sampling (2)

- In the frequency domain:

$$\text{comb}(t) \rightleftharpoons \text{comb}(f)$$

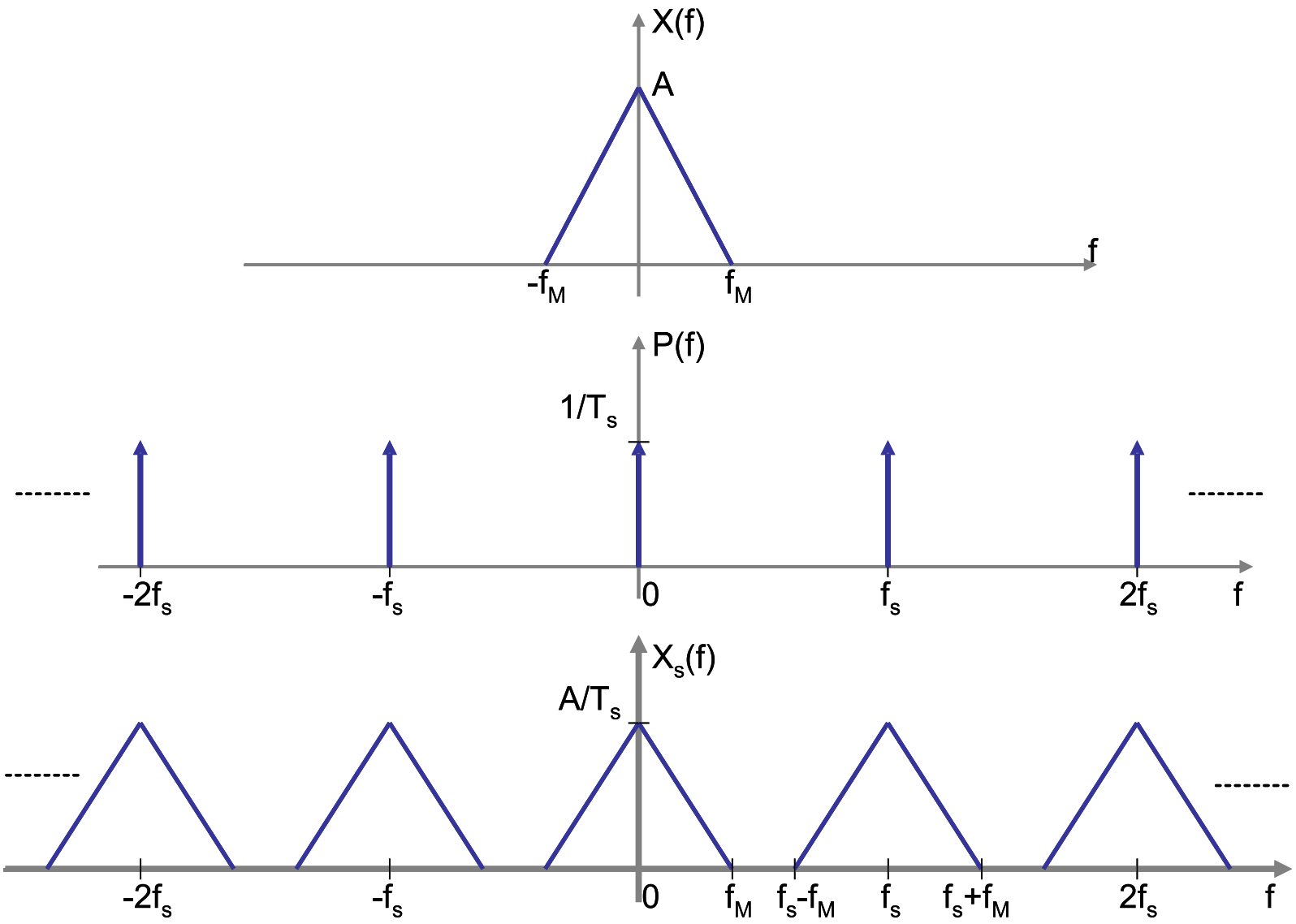
$$\text{comb}\left(\frac{t}{T_s}\right) \rightleftharpoons T_s\text{comb}(T_s f)$$

$$\frac{1}{T_s}\text{comb}\left(\frac{t}{T_s}\right) \rightleftharpoons \text{comb}(T_s f) = \frac{1}{T_s}\sum_{n\in\mathbb{Z}}\delta(f - nf_s)$$

- $x_s(t) = x(t)p(t)$ implies that:

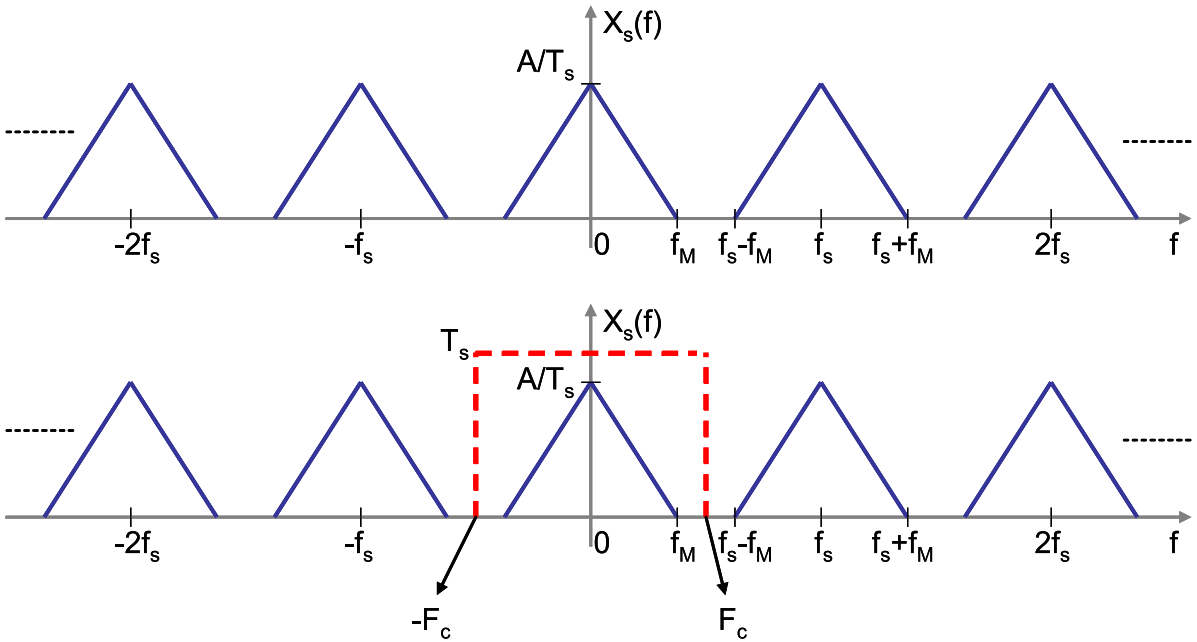
$$\begin{aligned} X_s(f) &= X(f) * P(f) \\ &= X(f) * \frac{1}{T_s}\sum_{n\in\mathbb{Z}}\delta(f - nf_s) \\ &= \frac{1}{T_s}\sum_{n\in\mathbb{Z}} X(f) * \delta(f - nf_s) \\ &= \frac{1}{T_s}\sum_{n\in\mathbb{Z}} X(f - nf_s) \end{aligned}$$

Impulse Sampling (3)



Impulse Sampling (4)

Reconstruction of $x(t)$:



- $x(t)$ can be reconstructed from the sampled signal:
  - If there is no overlap between the shifted spectra:

$$f_s - f_M \geq f_M \quad \Rightarrow \quad f_s \geq 2f_M$$

  - If the cutoff frequency of the lowpass filter verifies:

$$f_M \leq F_c \leq f_s - f_M$$

  - The gain of the filter must be equal to T_s.

Sampling Theorem

Consider a bandlimited signal $x(t)$:

$$X(f) = 0 \text{ for } |f| > f_M$$

Then $x(t)$ is uniquely determined by its samples $\{x(nT_s)\}$ if:

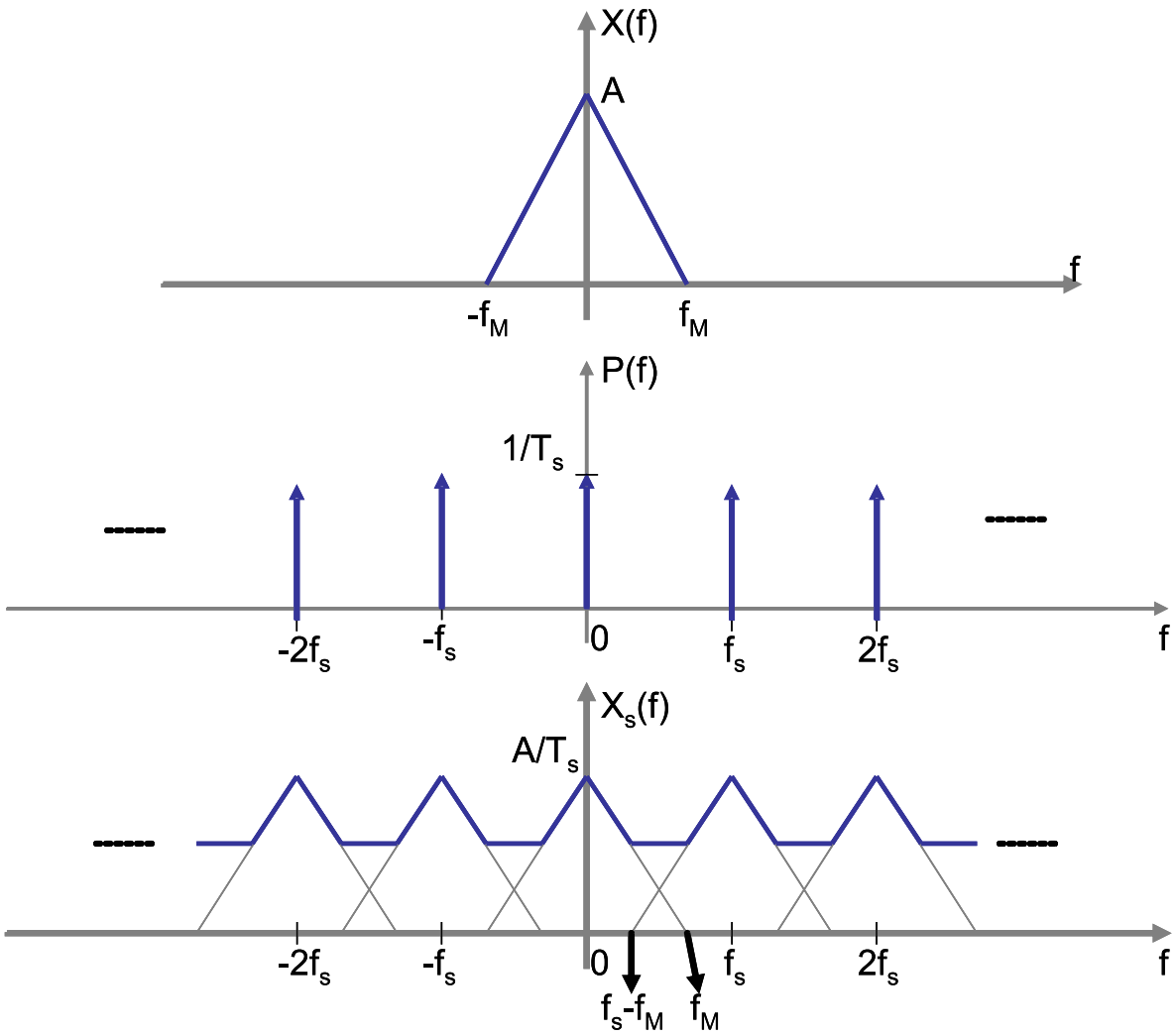
$$f_s \geq 2f_M$$

- The minimum sampling frequency should be $2f_M$ where f_M is the highest frequency in $X(f)$.

$f_s = 2f_M$ is called the Nyquist rate.

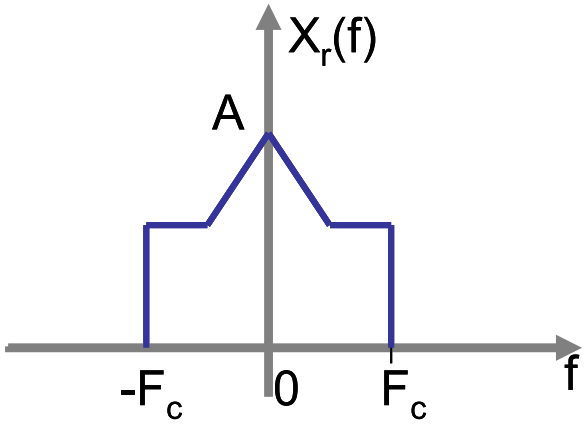
Undersampling and Aliasing (1)

Undersampling: The signal is sampled at a rate below the Nyquist rate.



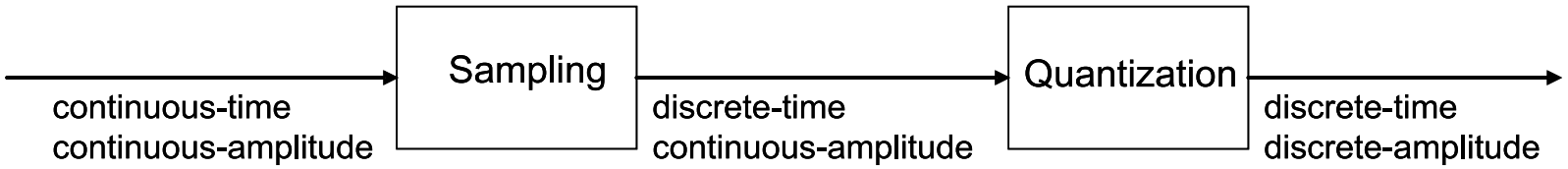
Undersampling and Aliasing (2)

After applying the lowpass filter having a gain T_s and a cutoff frequency F_c, the reconstituted signal will have the following shape:

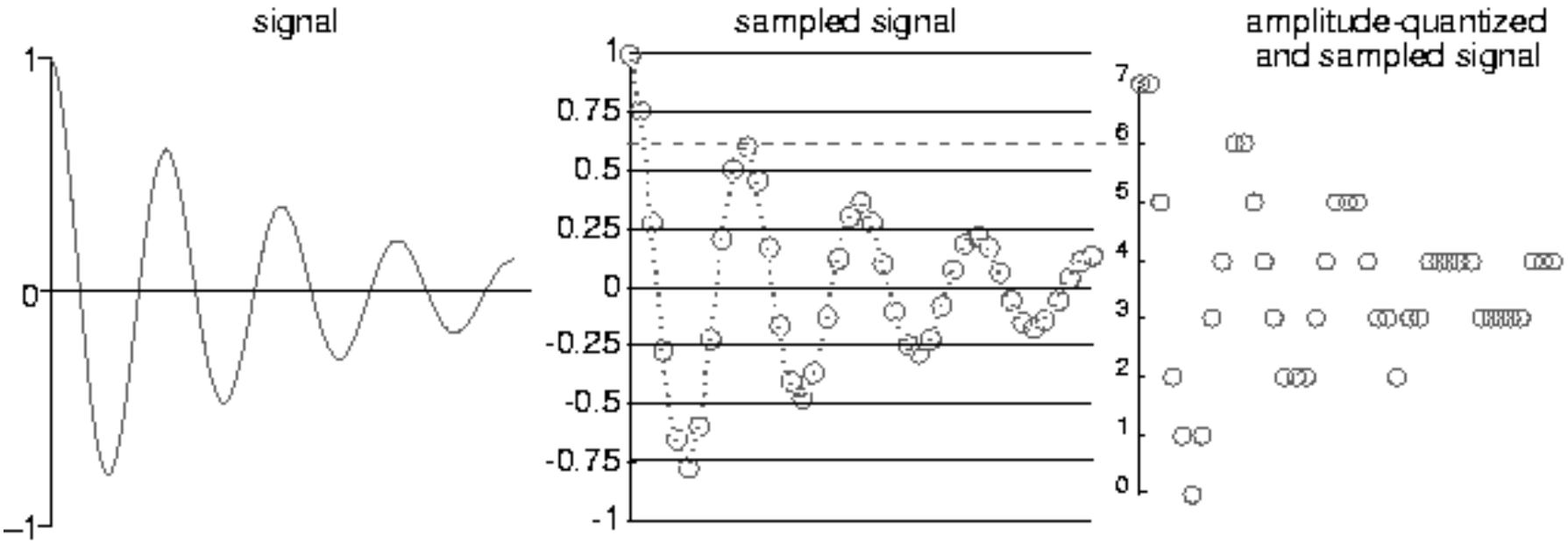


- Consequently, Distortions will be introduced because of the overlap between the different shifted spectra.
- This is known as aliasing.

Quantization (1)

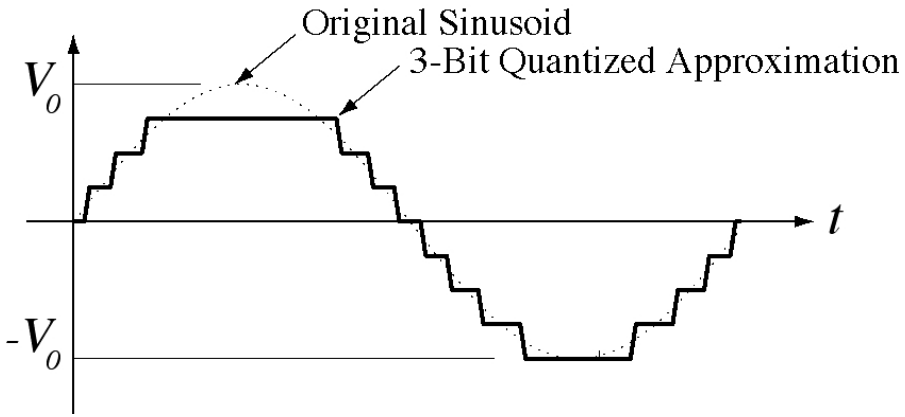


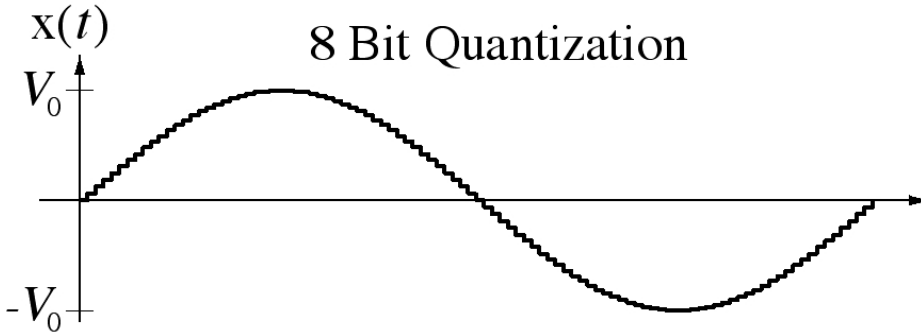
Quantization: The continuous range of amplitudes is approximated by a set of discrete values.



Quantization (2)

- In what follows, q will designate the number of quantization levels.
  - Example, in the previous figure, $q = 8$.
- In general, q is chosen to be a power of 2: $q = 2^n$. In this case, each quantization level can be mapped into a sequence of n bits.
  - Example, in the previous figure, $n = 3$ and the levels $0, \ldots, 7$ can be represented by the binary sequences $000, 001, \ldots, 111$.
- Note that quantization introduces errors.
  - Evidently, these errors decrease when q (or n) increases.





Quantization (3)

- We define the dynamic range D of $x(t)$ by:

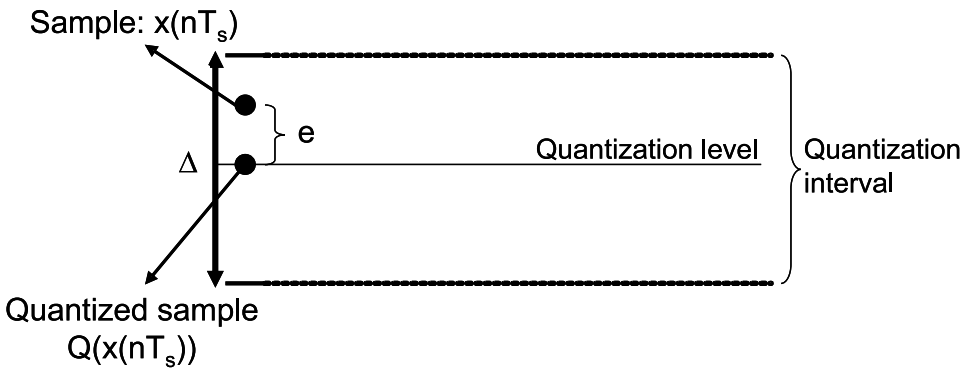
$$D = \max[x(t)] - \min[x(t)]$$

  - In our example, $D = 2$.
- Designate by Δ the width of the quantization interval:

$$\Delta = \frac{D}{q} = \frac{D}{2^n} = D2^{-n}$$

- When q increases, Δ decreases and $x(t)$ can be approximated by a linear function within each quantization interval.

Quantization (4)



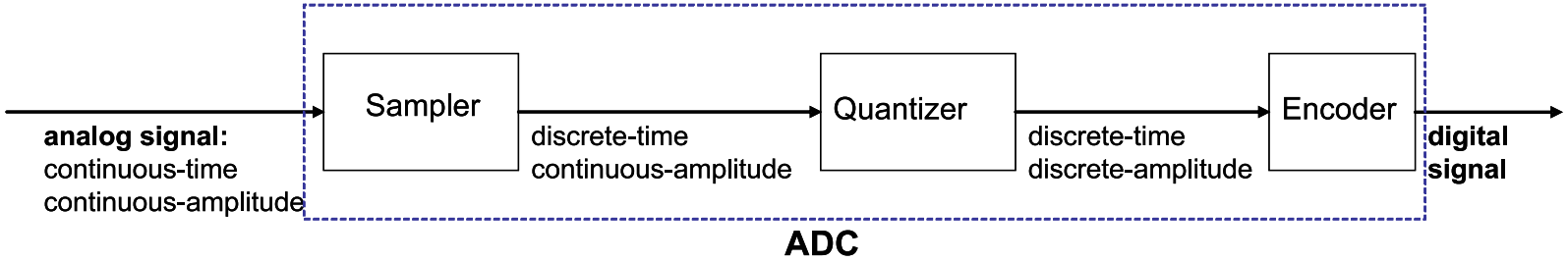
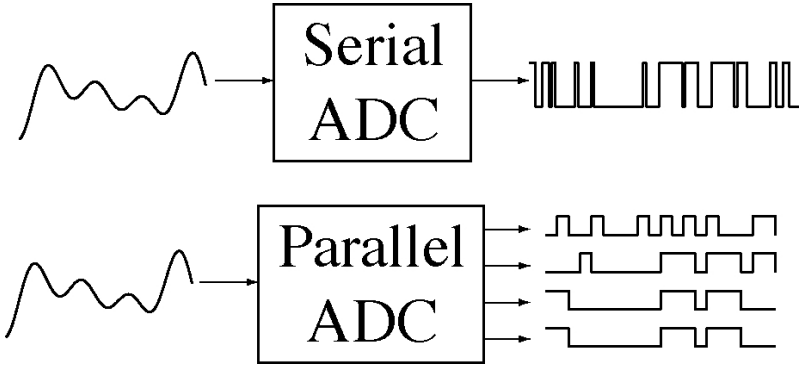
- The mean square error is given by:

$$E = \frac{1}{\Delta}\int_{-\frac{\Delta}{2}}^{+\frac{\Delta}{2}} e^2 \mathrm{d}e = \frac{\Delta^2}{12} = \frac{D^2 2^{-2n}}{12}$$

- Consequently, the quantization error decreases exponentially with the number of bits (n).
- Example, consider a compact disc (CD) audio signal:
  - This signal is sampled at 44 kHz $= 44,000$ samples/s.
  - Each sample is quantized over 2 bytes (16 bits) and therefore it can take $2^{16} = 65,536$ possible values.
  - In this case, the order of magnitude of the quantization error is $2^{-32} \approx 2.10^{-10}$.

Analog-to-Digital-Conversion

- Sampling and quantization are parts of the Analog-to-Digital-Conversion (ADC).
- The block diagram of an ADC is given by:



- The continuous-time signal is converted into a code. The code can be output serially or in parallel.